

SERENISUIT

Smart Physiological-Monitoring and Adaptive Calming Chest Suit Review 1 report — Demo 1, with the machine-learning component

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Review	Review 1 — Demo 1
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Signal modality	PPG (photoplethysmography) only
Revision	Supersedes the earlier ECG-based revision in full

What changed in this revision

The previous revision pursued two machine-learning tracks in parallel: arrhythmia classification from ECG, and arousal classification from PPG. That ECG track has been **retired in full**. It was trained on a public beat-level dataset whose rows cannot be attributed to individual patients, so its headline accuracy could not be defended as patient-independent, and it targeted a sensor (AD8232) that is not in the Demo 1 loop. Carrying a number that cannot survive scrutiny is worse than carrying a smaller number that can.

This revision is **PPG only**, which is also the only physiological signal Demo 1 actually acquires. The model is retrained on a larger dataset, validated leave-one-subject-out, exported to C at 1.3 KB and verified executing on the ESP32-S3. Every figure in this document was reproduced by re-running the committed training script before the report was written.

No circuit change was required. The classifier is pure firmware: no components, no wiring, no measurable current. The schematic, breadboard layout and bill of materials in the accompanying draw.io file are unchanged.

1. Demo 1 — what runs live

Demo 1 closes the sensing-to-feedback loop on real hardware. A fingertip is read by a MAX30102 optical pulse sensor over I²C. The ESP32-S3 detects beats, computes a rolling-average heart rate and displays it on a 0.96-inch SSD1306 OLED. If the rate falls outside 60-100 BPM the display shows DANGER and a vibration motor is driven through a DRV8833 H-bridge at 60 percent PWM duty, giving an average of about 3.0 V across a 3 V motor.

The live decision is a threshold, not the classifier — and that is deliberate. Section 2 gives the reasoning in full.

Scope note. The MAX30102 measures pulse by photoplethysmography. It reports heart rate, not blood pressure. Blood pressure requires a cuff or a calibrated pulse-transit-time arrangement and remains outside the scope of this prototype.

1.1 Hardware used in Demo 1

Component	Role	Interface
ESP32-S3-DevKitC-1 N8R8	Controller, 3.3 V logic, 8 MB flash	USB-C, host of all logic
MAX30102	Fingertip PPG	I ² C 0x57, 3.3 V

Component	Role	Interface
SSD1306 0.96" OLED	BPM and status display	I ² C 0x3C, 3.3 V
DRV8833	Motor driver, internal flyback diodes	GPIO4 PWM, VM 5 V
Vibration motor 3 V	Haptic alert	AOUT1 / AOUT2
100 μ F / 16 V capacitor	Absorbs motor inrush at VM	across VM and GND

Pin mapping is fixed by the DevKitC-1 J1 header: 3V3 on pin 1, GPIO4 on pin 4, GPIO8 (SDA) on pin 12, GPIO9 (SCL) on pin 15, 5V on pin 21 and GND on pin 22. Seating pin 1 at breadboard column 1 places every required pin in the matching column, all reachable from row A.

The AD8232 ECG module, GSR sensor, MPU-6050, flex sensor and LiPo supply are owned but deliberately left unconnected for Demo 1. The board is USB-powered throughout, which is the safer configuration while any biopotential electrode is out of the loop.

2. Why the live alert is a threshold and not the classifier

The two mechanisms answer different questions and must not be conflated:

	Threshold (live in Demo 1)	Classifier (validated, not live)
Detects	Abnormal heart RATE	Mental strain from the beat PATTERN
Input	One number: BPM	16 features over a 30 s window
Latency	A few seconds	30 s minimum, then 15 s steps
Needs calibration?	No	Yes — the wearer's own calm baseline
Determinism	Exact arithmetic	Probabilistic, 67.7 % accurate

A heart rate of 78 BPM is unremarkable and the threshold correctly stays silent. The same 78 BPM can nonetheless carry a beat-to-beat variability pattern typical of cognitive load, which the threshold is structurally blind to. That gap is the reason the classifier exists at all — a device that only asks whether BPM lies between 60 and 100 is a pulse monitor, not a stress-sensing suit.

Why it is nevertheless correct to keep it out of the live loop today. The classifier requires a calm baseline recorded from the wearer before it can score anything (Section 3.3). A baseline captured while the wearer is standing in front of a review panel is not a calm baseline; every subsequent window would be compared against an already-elevated reference, and the device would confidently report low strain in the most demanding moment available. Demonstrating a stress detector in that failure mode would misrepresent a result that is otherwise sound.

Demo 1 therefore demonstrates the classifier where the claim is exact and verifiable — that it trains, validates and **executes on the target microcontroller** — and leaves live scoring to Phase 2, where calibration can be performed properly. The threshold remains the live alert.

3. The machine-learning component

3.1 Dataset

Property	Value
Source	PPG Collection for Cognitive Strain (Kaggle)
Participants	22
Conditions	3-back task (high mental workload) vs 0-back (low)
Raw length	76,800 samples per trial = 300 s at 256 Hz
Windowing	30 s windows, 50 % overlap
Windows obtained	1672 (836 low / 836 high — perfectly balanced)
Median heart rate	79.1 BPM — sanity-checks the 256 Hz assumption

3.2 Features

Sixteen features are extracted per window rather than feeding raw samples, which are not time-aligned between subjects. After detrending and a 0.7-3.5 Hz band-pass (42-210 BPM), peaks are detected and converted to inter-beat intervals: mean heart rate, SDNN, RMSSD, pNN50 and the IBI coefficient of variation; log LF power, log HF power and the LF/HF ratio; and waveform statistics — amplitude standard deviation and range, mean absolute derivative, derivative standard deviation, peak mean and standard

deviation, skewness and kurtosis.

One correction worth recording. The LF (0.04-0.15 Hz) and HF (0.15-0.40 Hz) bands were initially computed by Welch PSD on the band-passed waveform. Those bands sit entirely below the 0.7 Hz high-pass corner, so the filter had already removed them and the three spectral features were near-meaningless. They are now computed from the IBI tachogram — the interval series interpolated onto a 4 Hz grid — which is the standard HRV construction. ROC-AUC improved from 0.748 to 0.7561 as a result.

3.3 Validation protocol — and the calibration question

Every window is tagged with its participant, and all reported figures use **leave-one-subject-out** cross-validation over 22 folds: the participant being tested never appears in training. Two protocols are compared, and the difference between them is the central finding of this work.

- **Protocol A — raw features.** The model is handed a stranger it has never seen and must judge them cold, with no personalisation.
- **Protocol B — wearer-calibrated.** Each participant's features are z-scored against that same participant's own low-workload windows, so the model sees deviation from the individual's own normal. This is what a wearable actually does: it is worn by one person and learns that person's baseline.

3.4 Results

Protocol	Model	Accuracy	ROC-AUC
A — raw (stranger)	Logistic Regression	0.5951	0.6232
A — raw (stranger)	Random Forest	0.5861	0.5879
A — raw (stranger)	XGBoost	0.5586	0.5729
B — wearer-calibrated	Logistic Regression (deployed)	0.6770	0.7561
B — wearer-calibrated	Random Forest	0.6687	0.7358
B — wearer-calibrated	XGBoost	0.6627	0.7277

Chance is 0.500 on perfectly balanced classes. Raw, subject-independent performance sits at 0.595 — barely above chance. Calibrating to the wearer lifts the same model to 0.677 with ROC-AUC 0.756, an unambiguous gain obtained without changing the features, the model or the validation protocol.

The finding: PPG encodes cognitive load only **relative to an individual's own baseline**. Absolute pulse statistics do not transfer between people, which is exactly why a stress-sensing garment must be calibrated to its wearer rather than shipped with a universal decision rule. The measurement above is the evidence for that design requirement.

Note also that the linear model wins on both protocols. With 16 features and 22 subjects, the tree ensembles have enough capacity to memorise individual participants and generalise worse for it, which is the same phenomenon quantified in Section 3.5.

3.5 The validation trap, quantified

If the windows are shuffled and split at random rather than by subject, windows from the same participant land on both sides of the split. The reported accuracy then rises sharply:

Model	Leave-one-subject-out (raw)	Random 5-fold (leaky)
Logistic Regression	0.5951	0.6262
Random Forest	0.5861	0.7907
XGBoost	0.5586	0.7775

Random Forest climbs from 0.586 to 0.791 — a gain of 20 points bought entirely by leakage. The tree models are recognising **who** the participant is, not how strained they are, and the linear model barely moves because it lacks the capacity to memorise identities. **These inflated figures are recorded here only to be excluded; they are not quoted as results anywhere in this project.**

3.6 Behaviour of the deployed model

	predicted low	predicted high
actually low	627	209
actually high	331	505

Specificity is 0.750 and sensitivity 0.604 (F1 0.652). The model is markedly better at recognising calm than strain, so its errors fall predominantly on missed strain rather than false alarms. For a garment whose response is to vibrate against the wearer's body that is the preferable direction of error: a missed episode is an absent prompt, whereas a false alarm is an unwanted physical interruption that itself raises arousal.

Feature importance (Random Forest, calibrated features):

Feature	Importance
hr	0.121
skew	0.085
peak_mean	0.080
kurt	0.075
amp_range	0.074
std_d	0.068

Mean heart rate leads, but the waveform-shape terms — skewness, peak amplitude, kurtosis, amplitude range — together outweigh it. That distribution is the quantitative argument that BPM alone is insufficient and the classifier is not redundant with the threshold.

4. On-device deployment

Logistic regression is the deployed model, and it needs no inference runtime at all. The trained standardiser and coefficients were exported directly as C arrays and the score computed as a dot product followed by a sigmoid — no TensorFlow Lite, no m2cgen, no external library.

Property	Value
Model	Logistic regression, 16 features
Exported as	src/ppg_model.c + include/ppg_model.h
C source size	1.3 KB
Share of 8 MB flash	about 0.02 %
RAM at inference	under 1 KB
Dependencies	none — plain C, no ML runtime

An export defect was found and fixed during this work: coefficients formatted with `%.6g` emitted bare `0f` for zero-valued terms, which is not a valid C float literal. Fixed-decimal formatting resolved it, and the generated source now compiles cleanly under `gcc -fsyntax-only`.

4.1 Self-test on the microcontroller

Eight real 30-second PPG windows — four low-workload, four high — are embedded in the firmware as calibrated feature vectors. On every boot the ESP32-S3 scores them and reports over serial at 115200 baud: the true label, the predicted label, P(high arousal) and the inference time in microseconds. It re-runs on demand by sending the character **m**.

This is a claim that can be checked live and cannot fail in front of an examiner, because it runs on stored data rather than on a nervous volunteer's finger. It establishes precisely one thing, and establishes it conclusively: **the model executes on the microcontroller**, in microseconds, within 1.3 KB — not in Python on a laptop.

5. What Demo 1 shows, and what it does not

- **Shows:** a complete sensing, decision and feedback loop on real hardware — a measured fingertip pulse driving a visible display and a haptic response.
- **Shows:** a classifier trained on 22 participants, validated leave-one-subject-out against three model families and two protocols, exported to 1.3 KB of C and executing on the ESP32-S3.
- **Shows:** a measured, reproducible argument for why wearer calibration is architecturally necessary.
- **Does not show:** live arousal classification. The classifier is not connected to the MAX30102 in this demo, and is not presented as being so.
- **Does not show:** ECG acquisition. The AD8232 is owned but out of the loop; the ECG model line has been retired rather than carried forward unvalidated.
- **Does not claim:** any diagnostic capability. This is a research and teaching prototype, not a medical device.

6. Phase 2 — the route to live classification

The remaining work is firmware, not hardware. No new component is required to make the classifier live.

- **Feature extraction on-device.** Dropping the three spectral features leaves 13 time-domain features and costs nothing measurable: accuracy 0.6770 — identical to the 16-feature model — with ROC-AUC 0.7492 against 0.7561. That removes the need for interpolation and a Welch periodogram on the MCU, so peak detection and simple statistics suffice.
- **A calibration mode.** The wearer rests for roughly 90 seconds while the device records their calm baseline, which is then stored and used to z-score every subsequent window.
- **Temporal agreement before acting.** At 68 percent per window, a single verdict is too noisy to drive a motor. Requiring two or three consecutive windows to agree suppresses most spurious triggers at the cost of 15-30 seconds of additional latency — a trade clearly worth making for a comfort device.
- **Two distinct haptic patterns.** An abnormal heart rate is an alarm and should feel urgent. High mental strain is the opposite: the intervention is biofeedback, so it should be a slow rhythmic pulse the wearer paces their breathing against. Buzzing someone sharply because they are stressed raises arousal rather than lowering it. Same motor, deliberately different waveforms.
- **Then multimodal fusion.** Add GSR, respiration and the MPU-6050 and repeat the experiment, testing whether fusion lifts performance beyond what PPG alone can reach.
- **Then the physical build.** Move from breadboard to soldered perfboard with JST connectors, then into the compression vest.

Appendix A — reproducing these numbers

```
cd demo1/ml
pip install scikit-learn pandas scipy matplotlib xgboost
python3 ppg_cognitive_strain.py

cd demo1/firmware
pio run -t upload
pio device monitor          # 115200 baud; press m to re-run the self-test
```

Appendix B — repository layout

serenisuit/		
demo1/		
diagrams/	SereniSuit_Demo1_Circuit.drawio	block diagram, schematic, wiring + safety, breadboard
firmware/	platformio.ini	ESP32-S3 build configuration
	src/main.cpp	live demo + ML self-test
	src/ppg_model.c	exported model, 1.3 KB
	include/ppg_model.h	
	include/ppg_test_windows.h	8 real windows for the self-test
ml/	ppg_cognitive_strain.py	training and evaluation
	dataset/cognitive_strain/	22 participants, High/Low MWL
report/	SereniSuit_Report_Demo1.pdf	this document